Name: _____

WORKSHEET

Circular calculations

1 A group of students sit at a circular table. Find the number of possible arrangements if there are:

a 5 students

b 9 students

c 11 students

d 13 students

e 15 students

f 20 students

2 **a** How many different ways can three people sit together (in a row)?

b Treating these three people who want to sit together as a single block, how many seats are available to sit at a round table that seats 10 people?

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c Hence find the number of possible arrangements for the table to seat 10 people where three particular people are seated together.

3 A round table seats 12 people. Find the number of possible arrangements if:

a 2 people want to sit together.

b 3 people want to sit together.

c 4 people want to sit together.

d 2 groups of 2 people want to sit together.

4 A round table seats 8 people. If an extra person is added to the table, how many *more* different possible arrangements are there?

5 How many different seats are at a round table if the number of different arrangements is:

a 720

b 120

c 6 227 020 800

d 39 916 800

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6 A 'Lazy Susan' in the centre of a table. On the 'Lazy Susan' there is a pair of salt and pepper shakers, three different gravies, two different dressings and two different sauces. What possible arrangements are there:

a when there are no restrictions?

b if the salt and pepper must be together?

c if the gravies must be together?

Answers

1 **a** 24

b 40 320

c 3 628 800

d 479 001 600

e 87 178 291 200

f 121 645 100 408 832 000

2 **a** $3! = 6$

b 8

c 30 240

3 **a** 79 833 600

b 21 772 800

c 8 709 120

d 1 451 520

4 35 280

5 **a** 7

b 6

c 14

d 12

6 **a** 40 320

b 10 080

c 4 320

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